



Find and Fix

Quick facilitation notes & answer key

Goal

Students diagnose and fix two programs with DIFFERENT kinds of bugs (a wrong-value block and an extra block), then name each bug type. Reading code, identifying the kind of error, and altering the code to correct it is the heart of C3.2, and naming the bug type builds debugging vocabulary.

Ontario curriculum (Grade 1 — C3 Coding)

C3.1 — solve problems and create computational representations of mathematical situations by writing and executing code, including code that involves sequential events.

C3.2 — read and alter existing code, including code that involves sequential events, and describe how changes to the code affect the outcomes.

Materials

The printed worksheet and a pencil. Optional: a floor number path 0-5 to run each program before fixing.

Run it (about 20 min)

1. I DO: run Program A — 'start 1, forward 1 to 2. Target is 4 — the forward block's number is too small.'
2. WE DO: Exercise 1 — A has a wrong-value block; change forward 1 to forward 3 (1 → 4). Exercise 2 — B overshoots to 4; it has an extra block; remove one forward 1 (0 → 2 → 3).
3. YOU DO: Exercise 3 — name the bugs: A = a wrong block (wrong number); B = an extra block.
4. Connect: 'Naming the kind of bug helps you pick the right fix.'

Answer key (modelled by the run-gate)

Ex 1 — Program A: 1 → forward 1 → 2 (misses 4). Bug = wrong value; change forward 1 to forward 3 so 1 → 4.

Ex 2 — Program B: 0 → 2 → 3 → 4 (misses 3). Bug = an extra block; remove one forward 1 so 0 → 2 → 3.

Ex 3 — Program A had a WRONG block (wrong number); Program B had an EXTRA block.

Watch for & differentiate

Common slip: trying to fix B by changing a number — the quickest fix is removing the extra step; accept any correct fix that lands on 3.

Simplify: tell students the bug type for each program and have them apply the matching fix.

Extend: ask students to make Program A wrong in a different way (an extra or missing block) for a partner to fix.

Success indicator: the child fixes A to reach 4 and B to reach 3 and names the two bug types correctly.